Sri Lankan Black Pepper

Maintenance and Crop Management

Knowledge Domain: 02 — Maintenance (Training, Pruning, Fertilization & Water Management)

This document provides structured knowledge on the maintenance and crop management practices for black pepper (Piper nigrum) cultivation in Sri Lanka. It covers vine training and shoot management, pruning techniques, shade and mulch control, fertilizer mixture composition, age-based application schedules (with and without Gliricidia lopping), and water management — all aligned with Department of Export Agriculture (DEA) guidelines.

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1. Training and Pruning of Pepper Vines

Proper training and pruning are the most critical crop management activities for black pepper. They determine the vine's shape, yield potential, and long-term productivity. Spikes — and therefore all yield — emerge only from lateral (plagiotropic) branches, making canopy architecture crucial.

1.1 Initial Vine Training

As soon as the young pepper plant begins to elongate after field planting, active training must begin:

- **Tying Method:** Tie the elongating vine to the Gliricidia support using a soft material such as cloth strips or soft twine. Never use wire or tight plastic ties — these can girdle and kill the stem.
- **Purpose of Tying:** Tying facilitates the development of adventitious roots that attach the vine to the support, creating a stable, permanent climbing structure.
- **Orthotropic Shoot Target:** Train 3 to 4 orthotropic (terminal / vertical) shoots up the support. These are the main framework branches of the vine.

1.2 Understanding Shoot Types

Black pepper produces two distinct types of shoots — understanding this is essential for correct pruning:

- **Orthotropic (Terminal) Shoots:** Orthotropic shoots grow vertically upward (terminal growth). They form the main structural framework of the vine. A vine should ideally have 3–4 orthotropic shoots climbing the support.
- **Plagiotropic (Lateral) Shoots:** Plagiotropic shoots grow horizontally as lateral branches. These are the productive branches — all fruiting spikes emerge exclusively from plagiotropic branches. A higher number of healthy lateral branches directly equals higher yield.

> **Note:** *Key Principle: Spikes emerge only from lateral (plagiotropic) branches — never from terminal orthotropic shoots. All pruning and training decisions should aim to maximise the number of healthy, well-lit lateral branches.*

1.3 Shoot Management at the 8–10 Node Stage

When the vine reaches 8 to 10 nodes, critical canopy management decisions must be made:

1. Assess whether 3–4 orthotropic shoots are developing alongside a satisfactory number of lateral plagiotropic branches.

2. If the vine is NOT producing orthotropic shoots at this stage, prune the vine at the terminal to stimulate the emergence of 3–4 new orthotropic shoots.
3. A vine with 2–3 strong terminal shoots forms a productive columnar canopy shape — the ideal form for high yield and light interception.
4. Allow lateral plagiotropic branches to develop freely from each node — these are the direct source of fruiting spikes and must not be removed unless diseased or damaged.

1.4 Canopy Height Management

- **Canopy Development:** After 3 to 5 years, the pepper vine grows to the top of the Gliricidia support and forms a full canopy.
- **Height Control:** At a height of 3.5–4.0 metres, prune the growing tips to stop further vertical extension and redirect energy into the lateral fruiting branches.
- **Harvesting Benefit:** Maintaining this height makes harvesting practical and concentrates yield within reachable range.

Full Pruning and Training Reference:

Task	When to Perform	Method / Notes
Initial Tying / Training	From first elongation	Tie young vine loosely to support using soft cloth — do not use wire or tight string
Orthotropic Shoot Selection	When vine reaches 8–10 nodes	Train 3–4 orthotropic (terminal) shoots up the support. Lateral plagiotropic branches develop from these
Pruning to Induce Orthotropic Shoots	8–10 nodal stage (if needed)	If the vine fails to produce orthotropic shoots, prune from the terminal to stimulate 3–4 new orthotropic shoots
Canopy Topping	When vine reaches 3.5–4.0 m	Prune terminal growth to maintain height and develop a productive columnar canopy shape
Side / Lateral Pruning	Year-round as needed	Remove dead, diseased, or overcrowded lateral branches to allow sunlight penetration and airflow
Gliricidia Lopping (Wet Zone)	4 times per year	Lop Gliricidia canopy to prevent overshading; apply lopped leaves at 10 kg/tree/year as mulch
Gliricidia Lopping (Other Zones)	3–4 times per year	Regulate height and branching to maintain support tree at 4–5 m; use cuttings as organic mulch

> **Note:** *What happens if you do NOT prune? Without topping, the vine grows uncontrollably tall — making harvesting impossible and concentrating all growth in unproductive terminal shoots. Without lateral pruning, the canopy becomes too dense and shaded, suppressing spike development, trapping humidity that encourages fungal disease, and dramatically reducing yield. Without Gliricidia lopping, excess shade suppresses flowering and fruiting. Neglecting pruning for even one season can take 1–2 years to correct.*

2. Shade Control and Gliricidia Management

Shade management is inseparably linked to productivity — too much shade from the Gliricidia support suppresses flowering, while too little exposes vines to heat stress. Gliricidia lopping is therefore both a shade control and organic fertilizer strategy.

2.1 Gliricidia Canopy Regulation

- **Target Height:** Regulate the height and branch density of the Gliricidia support to maintain a final canopy height of approximately 4–5 metres.
- **Lopping Frequency:** Prune Gliricidia at least 3 to 4 times per year. In the wet zone, four loppings per year is highly recommended as it reduces labour costs, controls excess shade, and provides a continuous supply of mulching material.
- **Lopped Material Use:** All lopped branches and leaves should be applied around the base of the pepper vine as green mulch.

2.2 Gliricidia Lopping as a Fertilizer Substitute

Experimental evidence from DEA field trials has confirmed that Gliricidia lopping, when applied at the correct rate, can significantly reduce chemical fertilizer requirements without any yield loss:

- **Application Rate:** Apply Gliricidia lopped material at a rate of 10 kg per tree per year around the base of the vine.
- **Fertilizer Saving:** This rate of Gliricidia lopping can cut the inorganic (chemical) fertilizer requirement by 50% — cutting both input cost and environmental impact.
- **Implication:** This is a core reason why the DEA provides two separate fertilizer schedules — one for plantations using Gliricidia lopping and one for those without.

> **Note:** *If Gliricidia lopping material is unavailable or insufficient, apply 16 kg of dried cow dung plus 12 kg of poultry manure mixture per vine per year as an organic alternative to achieve a similar soil enrichment effect.*

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3. Fertilizer Application

The DEA recommends a specific NPK + Kieserite fertilizer mixture for black pepper cultivation in Sri Lanka. Application rates vary based on vine age and whether Gliricidia lopping is being practised.

3.1 Recommended Fertilizer Mixture Composition

The standard fertilizer mixture is prepared by combining four components in specific parts by weight:

Fertilizer Component	Parts by Weight	Nutrient Contribution in Mixture
Urea (46% N)	4	14% N
Rock Phosphate (28% P_2O_5)	5	11% P_2O_5
Muriate of Potash (60% K_2O)	3	14% K_2O
Kieserite (24% MgO)	1	2% MgO

- **Full Rate:** Total mixture rate without Gliricidia lopping: 2,380 kg/ha/year.
- **Reduced Rate:** Total mixture rate with Gliricidia lopping (50% reduction): 1,190 kg/ha/year.

> **Note:** *Application method: Apply in a circular band approximately 30 cm away from the base of the vine and lightly incorporate into the topsoil to a depth of 2–3 cm using a hand hoe or rake. Always apply at the onset of the Maha (September–November) and Yala (March–May) monsoon seasons when soil moisture is high for maximum nutrient uptake.*

> **Note:** *Best time of day to apply: Apply fertilizer in the early morning (6–9 AM) or late afternoon (4–6 PM) — never during peak midday heat. If the soil is dry before the rains arrive, water the area lightly first and apply fertilizer once the soil is moist. This prevents fertilizer burn and ensures nutrients dissolve into the root zone. Never apply fertilizer to waterlogged or flooded soil.*

3.2 Age-Based Schedule — Without Gliricidia Lopping

Use this schedule when Gliricidia lopped material is NOT being applied (full fertilizer rate). Both kg/ha (plantation scale) and grams per vine (smallholder scale) are provided — based on the standard 1,700 vines/ha at 2.4 m × 2.4 m spacing:

Age of Plantation	Maha Season (kg/ha)	Yala Season (kg/ha)	Total per Year (kg/ha)
1st Year	250	250	**500**
2nd Year	500	500	**1,000**
3rd Year onwards	700	700	**1,400**

Per-vine equivalent (for smallholders — same data converted to grams per vine):

Vine Age	Maha (kg/ha)	Yala (kg/ha)	Per Vine — Maha	Per Vine — Yala	Total Per Vine / Year
1st Year	250	250	**147 g**	**147 g**	**294 g**
2nd Year	500	500	**294 g**	**294 g**	**588 g**
3rd Year onwards	700	700	**412 g**	**412 g**	**824 g**

3.3 Age-Based Schedule — With Gliricidia Lopping

Use this schedule when Gliricidia lopped material IS applied at 10 kg/tree/year (50% reduced fertilizer rate). Both scales provided:

Age of Plantation	Maha Season (kg/ha)	Yala Season (kg/ha)	Total per Year (kg/ha)
1st Year	125	125	**250**

Age of Plantation	Maha Season (kg/ha)	Yala Season (kg/ha)	Total per Year (kg/ha)
2nd Year	250	250	**500**
3rd Year onwards	350	350	**700**

Per-vine equivalent (for smallholders — same data converted to grams per vine):

Vine Age	Maha (kg/ha)	Yala (kg/ha)	Per Vine — Maha	Per Vine — Yala	Total Per Vine / Year
1st Year	125	125	**74 g**	**74 g**	**147 g**
2nd Year	250	250	**147 g**	**147 g**	**294 g**
3rd Year onwards	350	350	**206 g**	**206 g**	**412 g**

> **Note:** *For organic farming systems, substitute chemical fertilizers entirely with well-rotted cow manure or compost (5–10 kg per vine per year), supplemented with Dolomite (500 g per plant every two years) to maintain soil pH between 5.5 and 6.5.*

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4. Irrigation and Water Management

Adequate water management is vital during early establishment and during extended dry periods. Black pepper vines are sensitive to both drought stress and waterlogging.

4.1 Watering Guidelines

- **Young Plants (Years 1–2):** Provide regular watering during the first two dry seasons after planting — the root zone must never fully dry out during establishment.
- **Volume and Frequency:** Apply 5–10 litres of water per vine per watering session during dry spells. Water every 3–5 days during dry season for young plants, and every 7–10 days for established vines if rainfall is absent.
- **Best Time of Day:** Always water in the early morning (6–8 AM) or late afternoon (4–6 PM). Never water during peak midday heat — water evaporates rapidly and temperature shock can stress young roots.
- **Before or After Fertilizer:** If fertilizer is scheduled and soil is dry, water the vine lightly one day before applying fertilizer. This moistens the root zone for nutrient absorption. Do not apply fertilizer to waterlogged soil.
- **Mature Vines:** Mature vines (3+ years) rely primarily on monsoon rains. Supplemental irrigation is needed only during prolonged dry spells, particularly in dry zone districts.
- **Drainage Priority:** Ensure soil drainage is adequate at all times — waterlogging for more than 48 hours causes Phytophthora root rot, which can kill a vine within days.

4.2 Mulching Practices

Mulching is one of the most effective and low-cost water conservation techniques for black pepper:

- **Coverage Area:** Apply mulch around the base of the vine in a 50 cm radius to retain soil moisture and suppress weed growth.
- **Mulch Materials:** Recommended materials include dried straw, coconut husks, or freshly lopped Gliricidia leaves.
- **Frequency:** Replenish mulch at the start of each dry season. Over time, decomposing mulch enriches the topsoil with organic matter.

> **Note:** *Avoid placing mulch in direct contact with the vine stem — leave a 5–10 cm gap to prevent stem rot and fungal colonisation at the base.*

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5. Signs to Watch For — Field Diagnostic Guide

This section helps farmers and advisors identify exactly when action is needed on their pepper plantation. Each row describes a visible sign or trigger, the correct action to take, and how urgently to act. Use this table when a farmer says 'something looks wrong' or 'I'm not sure when to apply fertilizer / water / prune'.

Category	When / Season	Sign / Symptom to Look For	Recommended Action	Urgency
Pruning Needed		Vine has grown beyond 3.5–4 m with terminal shoots still growing upward	Top the vine — prune growing tips at 3.5–4 m to redirect energy to lateral fruiting branches	**Act within the season**
Pruning Needed		Vine at 8–10 nodes with no orthotropic (vertical) shoot development	Prune from the terminal immediately to stimulate 3–4 orthotropic shoots — this sets the vine's permanent structure	**Act immediately**
Pruning Needed		Dense overcrowded canopy — little light reaching inner lateral branches; increased fungal symptoms	Remove dead, weak, or overcrowded lateral branches. Lop Gliricidia if overshading is from the support tree	**Act within the month**
Pruning Needed		Gliricidia canopy has grown above 5 m or is casting full shade on vines below	Lop Gliricidia immediately. Maintain 4–5 m height. Apply all lopped material as mulch at vine base	**Act immediately**

Category	When / Season	Sign / Symptom to Look For	Recommended Action	Urgency
			(10 kg/tree/yr target)	
Fertilizer Needed		Maha season rains have begun — this is the scheduled fertilizer application window	Apply Maha dose in a circular band 30 cm from vine base, 2–3 cm deep. Water the area lightly before applying if soil is dry	**First 2 weeks of season**
Fertilizer Needed		Yala season rains have begun — this is the scheduled fertilizer application window	Apply Yala dose in a circular band 30 cm from vine base, 2–3 cm deep. Always apply when soil is moist — never into dry, cracked soil	**First 2 weeks of season**
Fertilizer Needed		Leaves are uniformly pale green or yellowing (not from disease spots or patterns)	Nitrogen deficiency likely — ensure Urea is correctly included in next fertilizer application. Do not over-apply — excess nitrogen causes lush foliage with poor yield	**Apply at next season**
Fertilizer Needed		Poor spike development, few flowers, or low berry set despite healthy vine growth	Phosphorus or Potassium shortfall likely — confirm Rock Phosphate and MOP are at correct ratio (5:3 by weight) in the mixture	**Apply at next season**
Water Needed		Topsoil (top 5 cm) is dry and cracking around the vine base	Apply 5–10 litres per vine directly to root zone. Water in early morning (6–8 AM) or late afternoon (4–6 PM) — never during peak midday sun	**Water within 24 hours**
Water Needed		Young vine leaves are wilting or rolling during midday — a sign of heat and moisture stress	Water immediately in early morning only. Apply 5 litres per plant. Increase mulch layer to conserve soil	**Water today**

Category	When / Season	Sign / Symptom to Look For	Recommended Action	Urgency
			moisture between waterings	
Water Needed		Mulch layer has thinned, dried out, or decomposed — vine base is exposed to direct sun	Replenish mulch immediately with straw, coconut husks, or Gliricidia cuttings. Cover 50 cm radius, leave 5–10 cm gap at stem	**Act within the week**
Drainage Problem		Standing water around vine base for more than 2 days; base of vine looks dark and wet persistently	Create drainage channels immediately. Phytophthora root rot sets in rapidly under waterlogged conditions. Do not irrigate further until soil drains	**Act immediately**

> **Note:** *Urgency colour guide: 'Act immediately' and 'Water today' = same-day action required. 'Act within the week/month' = schedule within that timeframe. 'Apply at next season' = note for the upcoming Maha or Yala application window.*